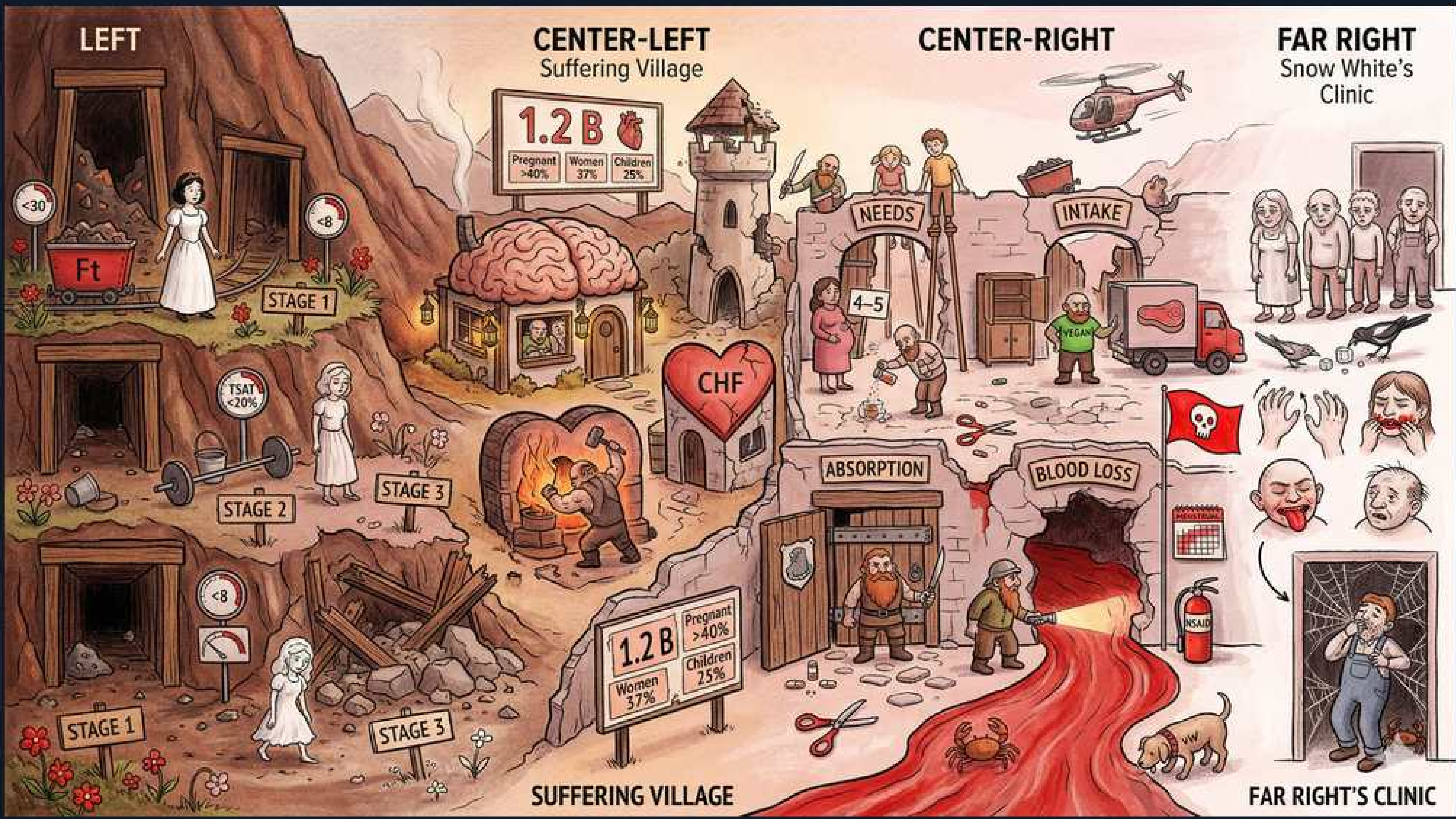
Iron / Microcytic — iron deficiency causes (Suffering Village)



1. Empty ferritin mine

WHAT YOU SEE

LEFT cave 'Ft' with pale Snow White, 'STAGE 1'

WHAT IT ENCODES

Stage 1 IDA = depleted iron stores. A LOW serum ferritin is the earliest and single most specific marker of iron deficiency: ferritin <30 ng/mL is highly specific, and <15 ng/mL is essentially diagnostic. Because ferritin is also an acute-phase reactant, in inflammation/infection/liver disease the cutoff to call iron deficiency rises (a value up to ~70–100 may still mean depleted stores).

BOARD TRAP

WRONG: thinking serum iron or Hb drops BEFORE ferritin. Sequence is stores first — ferritin falls earliest while serum iron, TSAT, and Hb are still normal. Ferritin is the first lab to move and the last to recover; do not wait for anemia to appear before calling iron deficiency.

---

2. Stage 2 — iron-deficient erythropoiesis

WHAT YOU SEE

Lower-left mine 'STAGE 2', weights/strain, low iron supply

WHAT IT ENCODES

Stage 2 = stores exhausted, so serum iron falls, TIBC (transferrin) rises, and TSAT drops below 20%, yet Hb is still near normal (iron-deficient erythropoiesis without overt anemia). Reticulocyte hemoglobin content (CHr/Ret-He) and ZPP begin to rise here before the CBC turns abnormal.

BOARD TRAP

WRONG: expecting microcytosis at this stage. MCV is still normal in stage 2 — microcytic, hypochromic indices are a late (stage 3) finding. A normal MCV does NOT exclude early iron deficiency; check ferritin/TSAT, not just red-cell size.

---

3. Stage 3 — frank anemia

WHAT YOU SEE

'STAGE 3' collapsed figure, microcytic cells

WHAT IT ENCODES

Stage 3 = overt microcytic (MCV <80 fL) hypochromic (low MCH/MCHC) anemia with HIGH RDW (anisocytosis) and low Hb. Smear shows pencil cells and target cells; platelets are often reactively elevated (thrombocytosis). This is the fully developed deficiency.

BOARD TRAP

WRONG: attributing a HIGH RDW automatically to iron deficiency only. RDW is high in IDA but NORMAL in thalassemia trait — RDW is the cheap clue that helps separate the two microcytic mimics (IDA = high RDW, thal = uniform small cells, normal/low RDW).

---

4. 1.2 B suffering / global burden

WHAT YOU SEE

Village sign '1.2 B' with pregnant 40%, women 37%, children 25%

WHAT IT ENCODES

IDA is the most common anemia and the most common nutritional deficiency worldwide (~1.2 billion affected). Prevalence is highest in pregnancy (~40%), menstruating women (~37%), and young children (~25%) because demand and/or loss exceeds intake during these windows.

BOARD TRAP

WRONG: assuming a microcytic anemia in a well-nourished Western adult must be dietary. In adults, dietary insufficiency is rare as a SOLE cause — blood loss dominates. Globally hookworm is a leading cause; in the US the answer is usually GI bleeding or menorrhagia, not diet.

---

5. Brain symptoms (pica/cognition)

WHAT YOU SEE

A brain-shaped hut in the village — the brain building IS the neuro/behavioral signs of iron deficiency: pica (ice-craving), restless legs, and impaired cognition.

WHAT IT ENCODES

IDA causes fatigue, restless legs syndrome, pica (craving ice = pagophagia, dirt, starch), koilonychia (spoon nails), angular cheilitis, glossitis, and impaired cognition/development in children. Plummer-Vinson syndrome = IDA + esophageal webs + dysphagia (a premalignant lesion for esophageal SCC).

BOARD TRAP

WRONG: dismissing restless legs or ice-craving as unrelated. Pagophagia is a near-specific symptom of iron deficiency and resolves with repletion; RLS should prompt a ferritin check (treat if ferritin <75).

---

6. CHF / cardiac strain

WHAT YOU SEE

Blacksmith forge labeled 'CHF'

WHAT IT ENCODES

Iron deficiency is independently associated with worse outcomes in heart failure with reduced EF. IV iron (ferric carboxymaltose) improves symptoms, exercise capacity, and reduces HF hospitalizations even WITHOUT overt anemia, using a functional cutoff: ferritin <100, OR ferritin 100–300 with TSAT <20%.

BOARD TRAP

WRONG: requiring frank anemia before treating iron deficiency in HF. In HFrEF the threshold is the FUNCTIONAL definition above; a normal Hb does not exclude treatable iron deficiency. Oral iron is ineffective in HF — use IV.

---

7. NEEDS / increased demand

WHAT YOU SEE

CENTER-RIGHT door 'NEEDS', figure '4–5'

WHAT IT ENCODES

Increased requirement outstrips intake: pregnancy (needs rise to ~4–5x baseline, ~27 mg/day recommended), lactation, infancy, and adolescent rapid growth. Erythropoietin therapy also creates functional iron demand that can outpace supply.

BOARD TRAP

WRONG: giving cow's milk to an iron-deficient toddler 'for nutrition.' Excessive cow's milk in toddlers is a classic cause of IDA — it is iron-poor, displaces iron-rich foods, and can cause occult GI bleeding. The fix is reducing milk and adding iron, not more milk.

---

8. INTAKE / poor diet

WHAT YOU SEE

Door 'INTAKE' with delivery truck

WHAT IT ENCODES

Inadequate dietary iron contributes in vegans/vegetarians (non-heme iron is poorly absorbed), the elderly, and food insecurity. Heme iron (red meat) is absorbed far better than non-heme (plant) iron; vitamin C boosts non-heme absorption.

BOARD TRAP

WRONG: in an adult, stopping at 'poor diet' as the final diagnosis. Diet may contribute, but unexplained IDA in an adult still mandates a search for blood loss — labeling it dietary and skipping GI evaluation misses colon cancer.

---

9. ABSORPTION / malabsorption

WHAT YOU SEE

Cave 'ABSORPTION' with small figures

WHAT IT ENCODES

Malabsorption causes: celiac disease (test anti-tTG IgA — iron is absorbed in the duodenum, the first part damaged in celiac), atrophic gastritis/autoimmune, H. pylori, gastric bypass/gastrectomy, and achlorhydria from chronic PPI use (acid is needed to reduce $Fe^{3+} \rightarrow Fe^{2+}$).

BOARD TRAP

WRONG: in refractory IDA that fails oral iron, assuming nonadherence and escalating dose. Forgetting CELIAC is the classic miss — IDA may be the only sign of celiac. Check anti-tTG IgA (with total IgA) before concluding the patient just isn't taking their pills.

---

10. BLOOD LOSS river

WHAT YOU SEE

Red river 'BLOOD LOSS' from cave with fire extinguisher, crab/scissors

WHAT IT ENCODES

Chronic blood loss is the #1 cause of IDA in adults. Each mL of blood ≈ 0.5 mg iron, so slow occult losses deplete stores over months. Sources: GI (peptic ulcer, colon cancer, angiodysplasia, hemorrhoids, NSAID gastritis) and gynecologic (menorrhagia).

BOARD TRAP

WRONG: skipping GI evaluation in a man or postmenopausal woman with IDA. Any IDA in these groups requires endoscopy AND colonoscopy to exclude GI malignancy — treating the anemia without finding the source can let a curable cancer progress.

---

11. Menorrhagia (women)

WHAT YOU SEE

FAR-RIGHT clinic, woman with heavy-flow imagery

WHAT IT ENCODES

Menorrhagia (heavy menstrual bleeding) is the leading cause of IDA in premenopausal women. Evaluate flow, consider fibroids/bleeding disorders (von Willebrand disease in young women), and treat both the anemia (iron) and the source (hormonal therapy, tranexamic acid).

BOARD TRAP

WRONG: attributing IDA to menses in a premenopausal woman without confirming the bleeding is actually heavy. If menstrual loss is normal, or anemia is out of proportion, GI sources still need consideration — don't reflexively blame the uterus and miss celiac or a GI bleed.

---

12. Occult GI bleed / colon Ca

WHAT YOU SEE

FAR RIGHT: spider web, gnawing animal, hidden bleeding

WHAT IT ENCODES

In men and postmenopausal women, unexplained IDA is colon cancer until proven otherwise — bidirectional endoscopy (EGD + colonoscopy) is mandatory. A right-sided colon cancer classically presents as iron-deficiency anemia with little overt bleeding.

BOARD TRAP

WRONG: attributing IDA in an older man to 'diet' or hemorrhoids and stopping there. Hemorrhoids do not explain IDA and must not be accepted as the cause until proximal lesions are excluded by colonoscopy; right-colon cancers bleed occultly and present as IDA.
